

SERVICE MANUAL



Document 780029

Revision B

**GE Signa[®] 1.0T
FLOW 7000
PHASED ARRAY PERIPHERAL VASCULAR COIL**

GE Catalog Part Number: M1887PW

USAI Part Number: 160068

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DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

Immediately complete a "Damage Loss Claim Form", available via MS Exchange Mail, after the damage is found.

MS Exchange Path:

Outlook/Public Folder/All Public Folders/Medical Systems/!Global Initiatives/Information Management/Forms/Common Forms/DAMAGE LOSS CLAIM FORM.

Send the completed form to the email address listed in the form.

For more information about the Transportation Claim Procedure, access the GE Medical Systems Intranet and enter the following URL address (case sensitive):

<ftp://3.87.40.2/globepro/qualsys/Docs/190016MF.PDF>

Language Policy For Service Documentation (Dir. 2128126)

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SECTION 1 – INTRODUCTION

1-1 Product Identification and Shipping List

To identify the Flow 7000 Phased Array Peripheral Vascular Coil, refer to the coil label (as shown in *Figure 1*).

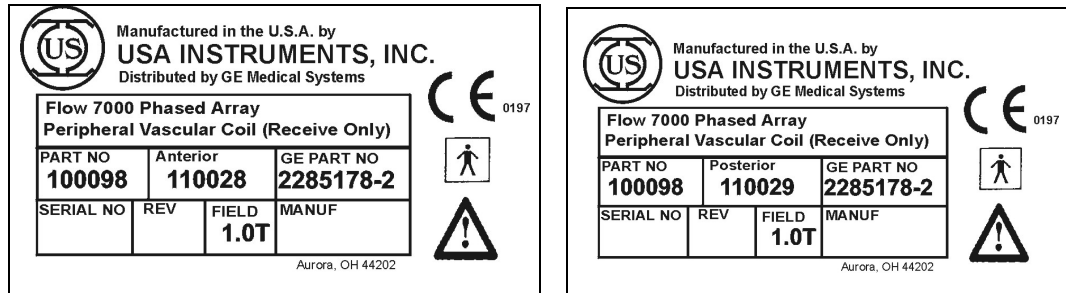


Figure 1: Coil labels.

Figure 2 shows a picture of the Flow 7000 Phased Array Peripheral Vascular Coil.



Figure 2: The Phased Array Peripheral Vascular Coil.

SHIPPING LIST – TABLE 1-1

Box #	Part Name	GE Part #	USAI Part #	Qty
1	Coil	2285178-2	100098	1
1	Ramp Pad	E8800ZT	150040	1
1	Patient Comfort Pad	E8800ZW	150075	1
1	Foam Spacer Block	E8800ZY	150041	1
1	Velcro Strap Set (attached to coil)	E8800ZZ	150042	1
1	Phantom Strap Set	2285178-4	150171	1
1	Operator's Guide	2285178-5	770016	1
1	Service Manual	2285178-12	780029	1
2	Phantom Positioner Set	2192406-9	150036	1
3	Loading Phantom	2192406-7	150035	2
3	SNR Phantom	2192406-6	150034	1
4	SNR Phantom	2192406-6	150034	2

1-2 Compatibility

The coil is compatible with the Signa[®] Horizon[™] and LX Horizon[™] 1.0T.

1-3 Related Documentation

Operator's Guide, GE Part Number 2285178-5.

Signa[®] LX Service Methods CD, 2160623-1.

1-4 Environmental Requirements

Storage Requirements

To store the coil, a storage space of 65.5" x 27.87" x 12" is required. Lay the ramp pad and posterior array coil completely flat. Place a sheet on top of the posterior array coil. The sheet separates the posterior array coil and anterior array coil to prevent them from sticking. Place the anterior array coil on top of the sheet and loosely fasten the Velcro straps. Note that although the straps are fastened together, the coil is not bent.

Note: The coil should be handled with care. Gently bend the coil when necessary – never fold/crease the coil sharply. Avoid excessive folding of the coil.

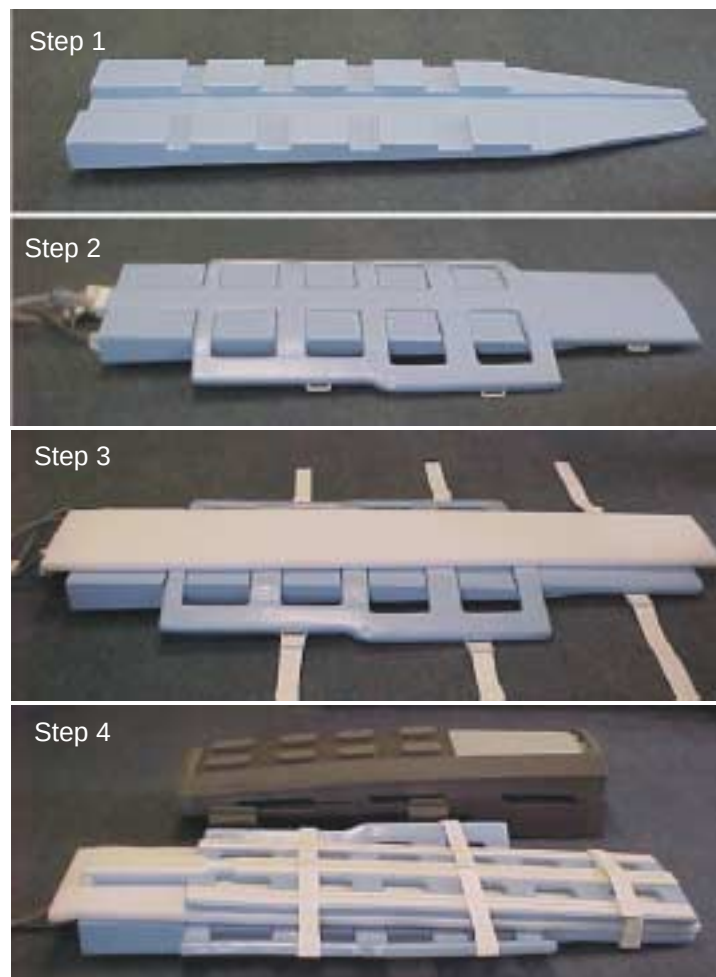


Figure 3: Storing the coil.

The phantoms may be stored next to the coil as shown in *Figure 3* or in the shipping box as shown in *Figure 4*.



Figure 4: Storing the phantoms in the shipping box.

Dimensions

Ramp Pad	65.48" x 15.50" x 3.65" (166.3 cm x 39.37 x 9.27 cm)
Posterior Coil	65.50" x 27.87" x 2.13" (166.4 cm x 70.79 cm x 3.99 cm)
Anterior Coil	65.34" x 14.50" x 1.62" (166.0 cm x 36.83 cm x 4.11 cm)
Loading Phantom	17.30" x 9.4" x 1.6" (44.0 cm x 23.8 cm x 4.1 cm)
SNR Phantom	Diameter: 6.6" (16.8 cm), Length: 13.8" (35.0 cm)

Weight

Ramp Pad	7 lbs. (3.175 kg)
Posterior Coil	20 lbs. (9.072 kg)
Anterior Coil	15 lbs. (6.804 kg)
Loading Phantom	11 lbs. (4.99 kg)
SNR Phantom	19 lbs. (8.60 kg)

1-5 Theory of Operation

The coil package is a flexible two-piece design. The coil also has a ramp pad that is positioned underneath the lower coil piece. The purpose of the ramp pad is to facilitate the positioning of the body vasculature in a horizontal plane. The coil contains 12 elements that are physically arranged into six pairs. A block diagram showing individual coil elements of the Phased Array Peripheral Vascular Coil is shown in *Figure 5*. A maximum of four RF inputs, representing either the upper (PVUPPER), middle (PVMIDDLE) or lower (PVLOWER) four coils can be selected at a time (see *Figures 6, 7 and 8*).

The Flow 7000 Phased Array Peripheral Vascular Coil is a receive-only coil. The coil is decoupled from the transmit coil during transmit by means of a RF choking circuit. These RF chokes can be switched on actively or passively. Each loop coil has six decoupling RF chokes (one active and five passive) and each saddle has eight decoupling chokes (one active and seven passive). Passive decoupling is achieved when small signal diodes are turned on by the induced RF voltage, in the chokes, coupled from the transmit field. Each small signal diode is turned on when the induced RF voltage reaches 0.5 volt. Active decoupling is achieved by PIN diodes that are turned on by a biasing DC current source (200mA) from the MR scanner. When either the small signal diodes or PIN diodes are turned on the RF choking circuits become very high impedance (typically above 2 kilo-ohm) blocks, compared to the other circuit elements (less than 25 ohm). These high impedance elements segregate the coil circuitry into several electrical segments.

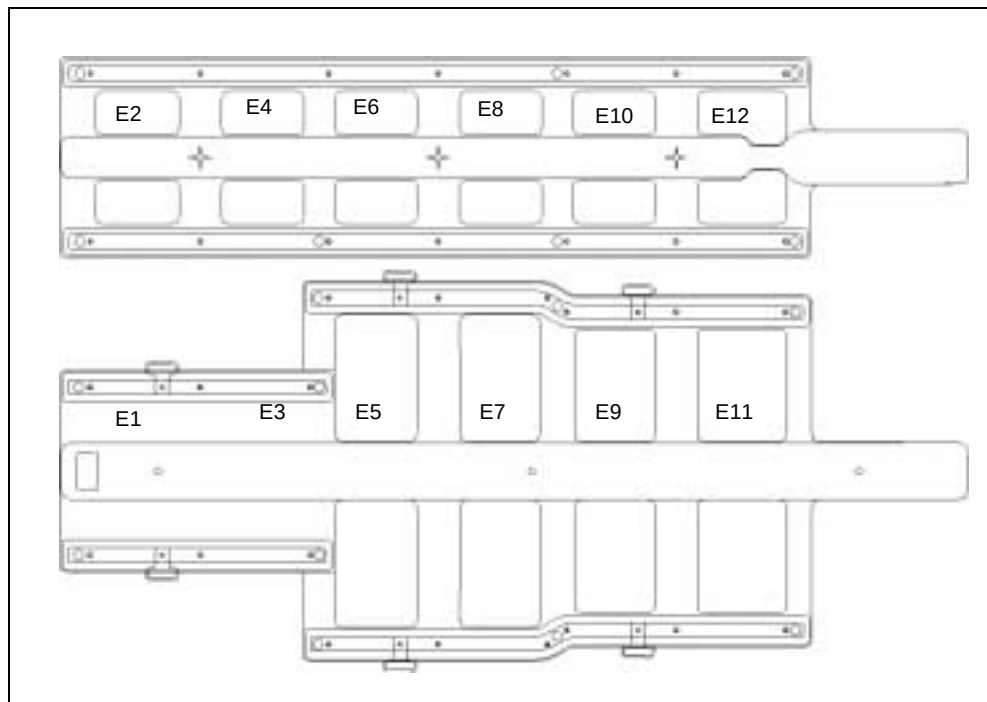


Figure 5: Block diagram of the coil.

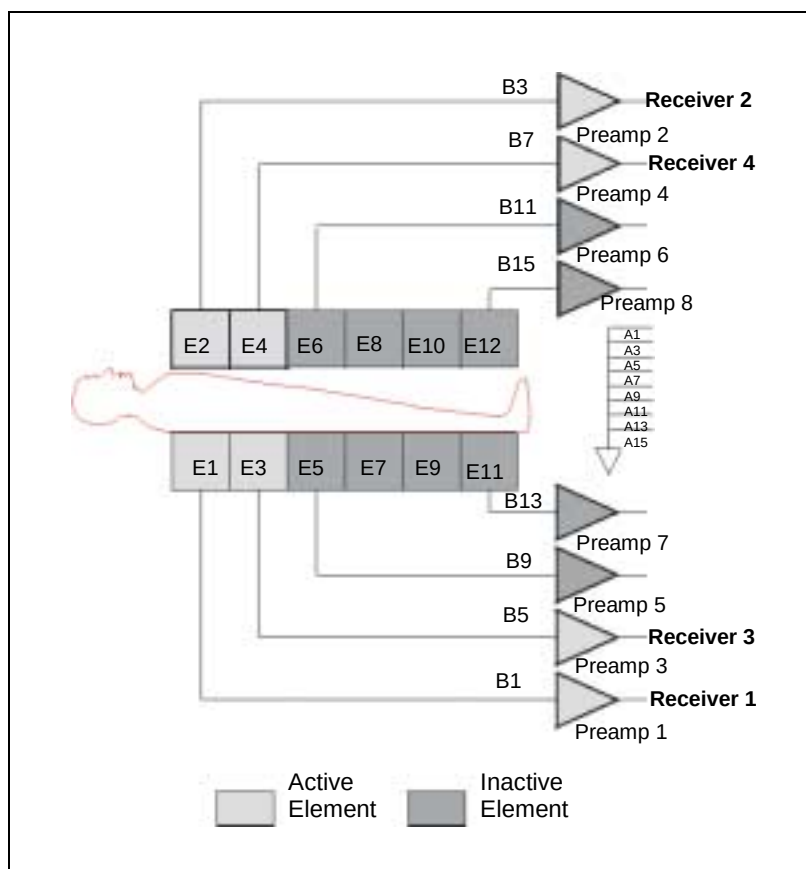


Figure 6: "PVUPPER" signal path (port mask 3).*

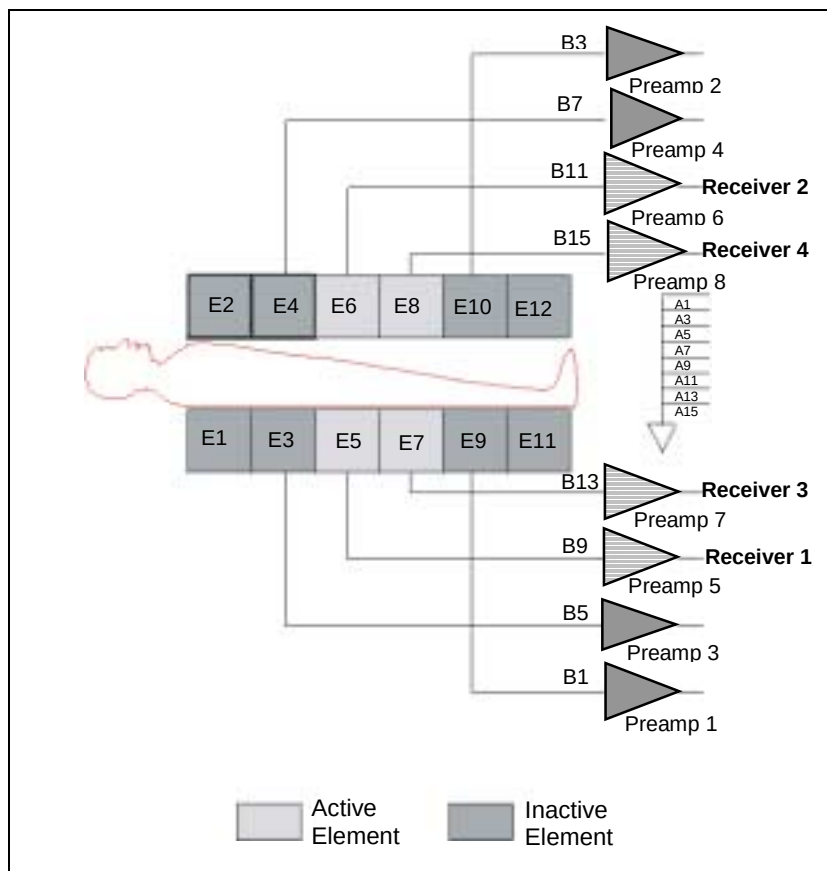


Figure 7: "PVMIDDLE" signal path (port mask 12).*

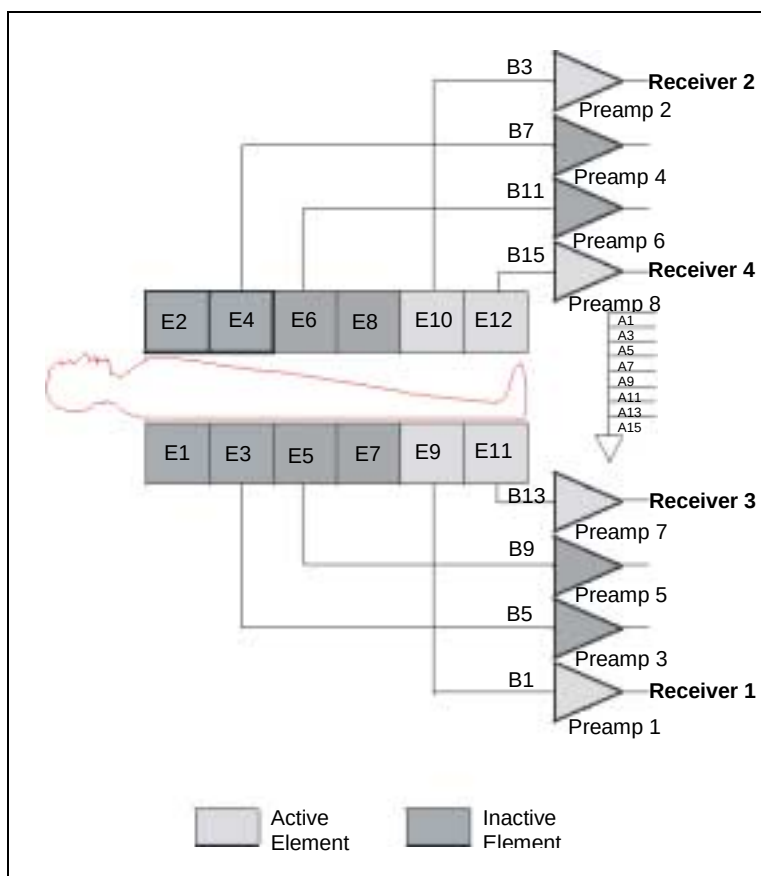


Figure 8: "PVLOWER" signal path (port mask 9).*

*Note: LX and Genesis Systems Receiver Numbers – Table 1-5 relates receiver numbering in LX and Genesis systems.

LX AND GENESIS SYSTEMS RECEIVER NUMBERS – TABLE 1-5

System	Receiver Numbers			
LX User Interface	1	2	3	4
LX Hardware Docs	0	1	2	3
Genesis User Interface	0	1	2	3
Genesis Hardware Docs	0	1	2	3

SECTION 2 – SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

Installing the Preamplifiers

The Phased Array Peripheral Vascular Coil requires all eight (8) phased array preamplifiers to be installed in the system. Refer to the system manuals for information on installing the preamplifiers. Also, refer to instructions in the preamplifier kit (GE Part Number M1887CY).

If the coil will be used on more than one system, a preamplifier kit for each system must be installed.

Installing the Coil

The name for this coil is: PV Array.

Add the coil using the Configuration File Manager. Refer to: Service Methods CD; System Level Procedures; Software Utilities.

If the coil does not exist in the Coil Config File, refer to the “Adding New Coils to Config File Manager” procedure and use the coil configuration information in Section 7-3 of this manual.

2-2 Installation Functional Checks

1. Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.
2. Perform Section 3-2 Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks as outlined below to ensure the coil is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3-2 Coil Imaging Performance Verification and record the data values on the Data Sheet.

SECTION 3 – FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-2-1

Description	GE Part #	USAI Part #	Qty
SNR Phantom	2162406-6	150034	3
Loading Phantom	2192406-7	150035	2
Phantom Positioner Set	2192406-9	150036	1
Phantom Strap Set	2285178-4	150171	1

3-2-2 Explanation of Procedure

The PV array has three sections controlled by the coil configuration files: PVUPPER, PVMIDDLE, and PVLOWER. For evaluation of coil performance, sagittal images will be acquired on the three sections of the coil using the protocol given below.

SNR measurements must be made for each mode, requiring three sets of signal and noise scans. Refer to the Data Sheet in Appendix 7-1 to understand the data required to calculate the individual element SNR for each mode of operation. All ROI measurements are made on the individual element images, **not** on the composite image.

The image quality check uses two different protocols for signal and noise image acquisition. The signal scan is an **FSE** sequence used to minimize susceptibility and B₀ inhomogeneity effects. The noise scan is a **GRE** sequence that has a Control Variable (do_noise) to eliminate the transmit RF completely during the scan. The signal scan **must** be run prior to the noise scan as the R1, R2, and TG values from the signal scan are used for the noise scan.

3-2-3 Signal Scan

1. From the Scan Desktop, start new scan by selecting [New Pt]; set **Patient ID** to “geservice” and **Patient Weight** to “111” pounds. Click [Patient Position] to open protocols window.
2. Place the ramp pad, the posterior array coil, the phantom positioner lower section, and the loading phantom on the patient table as shown in *Figure 10*. The coil should be positioned such that its cables exit toward the magnet.
3. Position the three SNR phantoms in the phantom positioner lower section so the handles do not interfere with the phantom positioner upper section. Place the phantom positioner upper section with the loading phantom over the SNR phantoms as shown in *Figure 10*.

4. Lastly, place the anterior array coil over the phantom positioner upper section and loading phantom, and secure them together with the Velcro straps as shown in *Figure 10*.

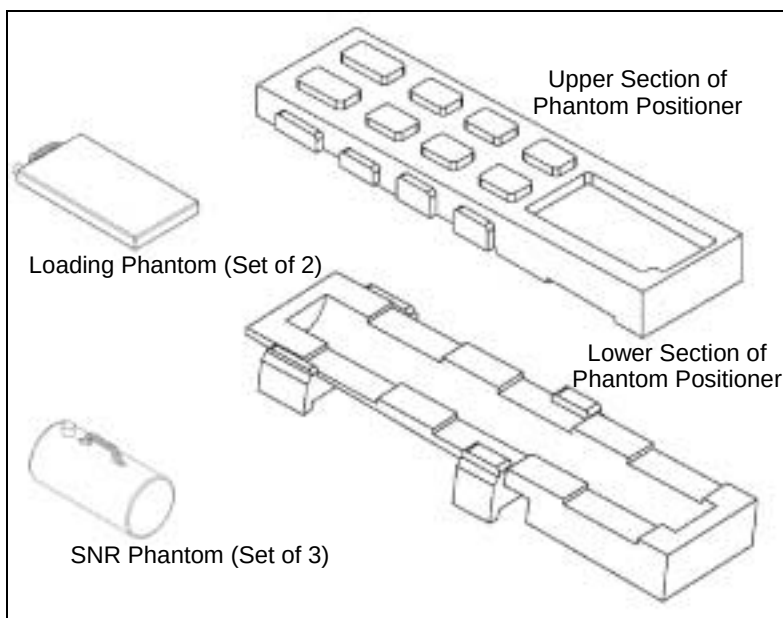


Figure 9: Parts of the coil phantom set.

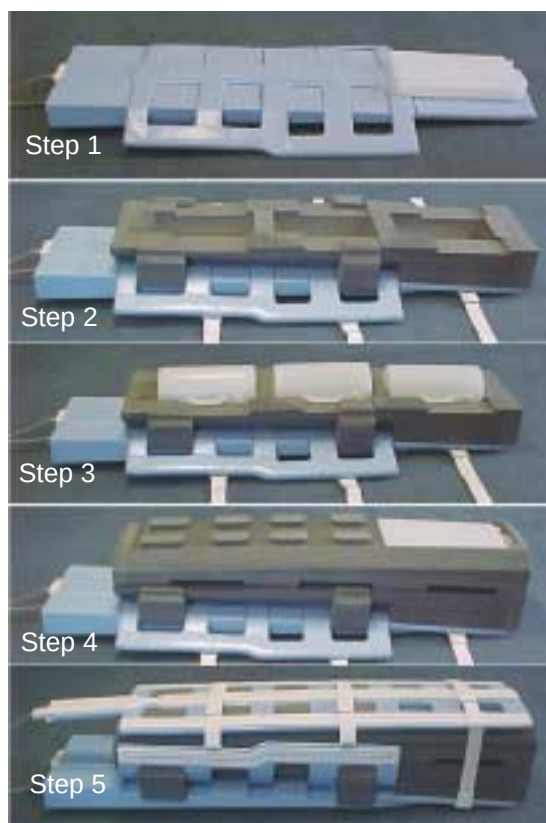


Figure 10: Assembling the coil phantom set.

5. For the Upper section, set the landmark on the landmark line that is marked on the upper (head) end of the coil (see *Figure 11*). When scanning the Middle section, set the landmark in the middle landmark line (see *Figure 11*). For the Lower section, set the landmark on the lower landmark line (foot) end of the coil (see *Figure 11*).

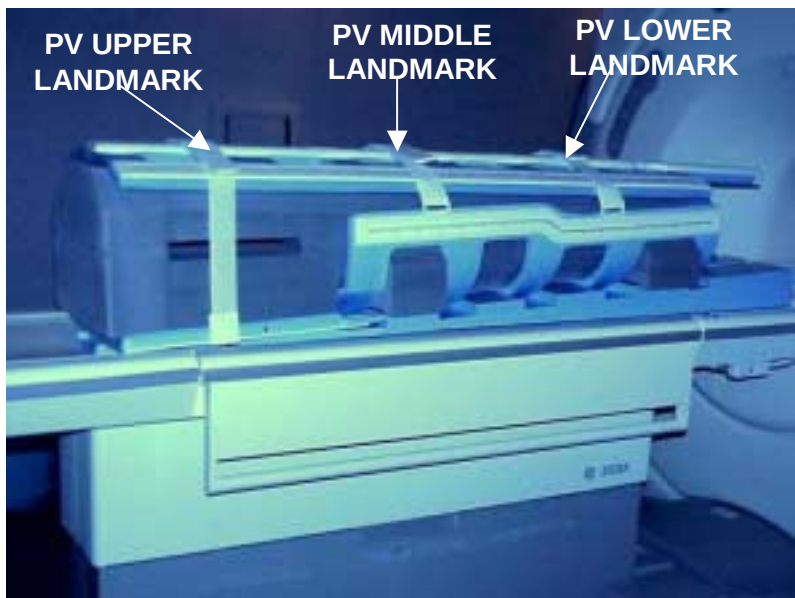


Figure 11: Coil landmarks.

6. At the magnet, press the “**Alignment Light**” button to turn on the light. Move the cradle to align the coil to the alignment lights as shown in *Figure 11*. Press the “**Landmark**” button to landmark the alignment.
7. Move the coil to scan position by pushing the “**Move to Scan**” button, ensuring the cable does not get snagged.
8. At the operator console, set the protocols per the Signal section from Signal and Noise Protocols – Table 3-2-4.
9. Click [**Save Series**] to download the protocols; then click [**Prepare to Scan**].
10. Open the [**Display CVs**] menu under [**Research Operations**] (click the right mouse button). Set the “**saveinter**” CV to “1” (saves the intermediate images so ROI measurements can be performed).
11. Run [**Auto Prescan**]. Record the R1, R2 and TG values on the SNR Data Sheet (found in the Appendix).
12. Run [**Scan**].

3-2-4 Noise Scan

A signal scan must be run **prior** to the noise scan as the same R1, R2 and TG values must be used for both the signal and noise scans. Do **not** run an Auto Prescan prior to the noise scan as the values will be changed.

1. Copy the signal scan series. Use [**Copy Series**] (highlight signal series and click the right mouse button) and [**Paste Series**] in **RX Manager**.
2. Click [**View Edit**] and set the protocols per the Noise section from Signal and Noise Protocols – Table 3-2-4.
3. Click [**Save Series**] and click [**Prepare to Scan**].
4. Open the [**Display CVs**] menu under [**Research Operations**]. Set the “**saveinter**”, “**rhformat**”, and “**do_noise**” CVs to “1”.
5. Run [**Manual Prescan**], do **not** make any changes, and click [**Done**].
6. Run [**Scan**].

SIGNAL AND NOISE PROTOCOLS – TABLE 3-2-4

Protocol	Signal	Noise
Patient/Exam Information		
Patient ID	geservice	geservice
Patient Name	PV Array	PV Array
Patient Weight	111 lbs. (50 kg)	111 lbs. (50 kg)
Patient Position		
Patient Position	Supine	Supine
Patient Entry	Feet First	Feet First
Coil	CPVUPPER, CPVMIDDLE, or CPVLOWER	CPVUPPER, CPVMIDDLE, or CPVLOWER
Series Description	Signal	Noise
Imaging Parameters		
Plane	Sagittal	Sagittal
Mode	2D	2D
Pulse Seq	FSE	GRE
Imaging Options	Variable Bandwidth	Variable Bandwidth
PSD Name	<i>leave blank</i>	<i>leave blank</i>
Protocol	<i>leave blank</i>	<i>leave blank</i>
Scan Timing		
# of Echoes	1	1
TE	17	17
TR	500	500
Echo Train Length/Flip Angle	4	1
Bandwidth	15.63	15.63
Additional Parameters		
<i>no entries required in this area</i>		
Acquisition Timing		
Freq	256	256
Phase	256	256
NEX	1	1
Phase FOV	1	1
Freq DIR	S/I	S/I
Auto Center Freq	Peak	Peak
Autoshim	On	On
Phase Correct	On	On
Contrast	Off	Off
# of Reps B4 Pause	0	0
Scanning Range		
FOV	38	38
Slice Thickness	3	3
Spacing	1.5	1.5
Start R/L	0	0
P/A Center	A50	A50
I/S Center	0	0
End R/L	0	0
Slices	1	1
Table Delta	0	0

3-2-5 SNR Image Analysis

SNR Measurement

Regions of interest in both signal and noise images can be measured directly in the image browser. Click the user interface button Measure, select the circular or rectangular shape, and adjust its size and orientation when the shape is displayed in the selected image. Mean, standard deviation, and area of the ROI will appear in the lower right corner of the image.

For the signal measurement, choose a rectangular ROI covering the appropriate section of the phantom for the receiver channel being scanned. The rectangular ROI should be about 8600 mm². ROIs are shown in *Figures 12 through 16*.

The SNR is calculated using the signal to noise ratios of the individual receiver channels:

$$\text{SNR}_{\text{Element}} = \frac{\text{Mean of Signal within ROI}}{\text{Standard Deviation of Noise within ROI}}$$

Note: The SNR calculation uses the **MEAN** of the signal image and **STANDARD DEVIATION** of the noise image. SNR is measured for each **element**, not on the composite image.

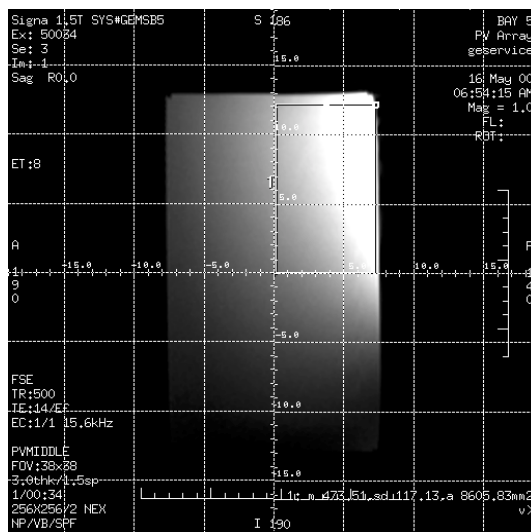


Figure 12: PV Upper/Middle/Lower Receiver 1.
ROI $\approx 8600 \text{ mm}^2$.
7.5 cm A-P, 12.5 cm S-I.

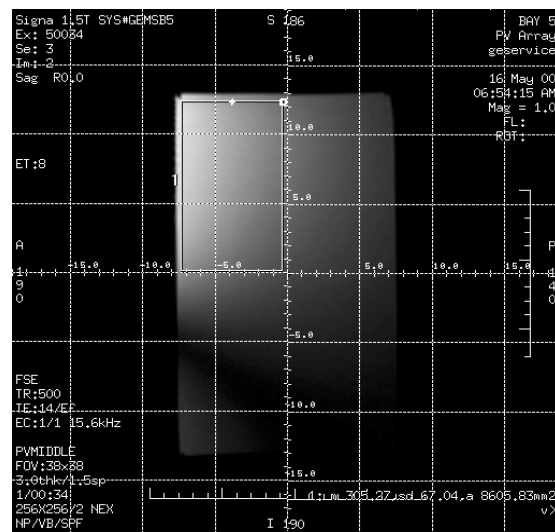


Figure 13: PV Upper/Middle/Lower Receiver 2.
ROI $\approx 8600 \text{ mm}^2$.
7.5 cm A-P, 12.5 cm S-I.

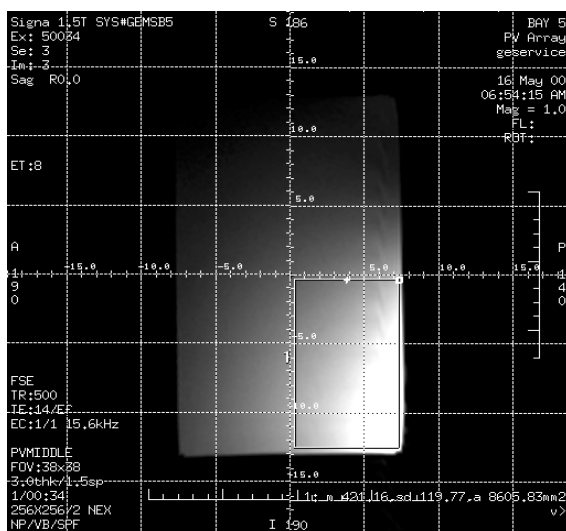


Figure 14: PV Upper/Middle/Lower Receiver 3.
ROI $\approx 8600 \text{ mm}^2$.
7.5 cm A-P, 12.5 cm S-I.

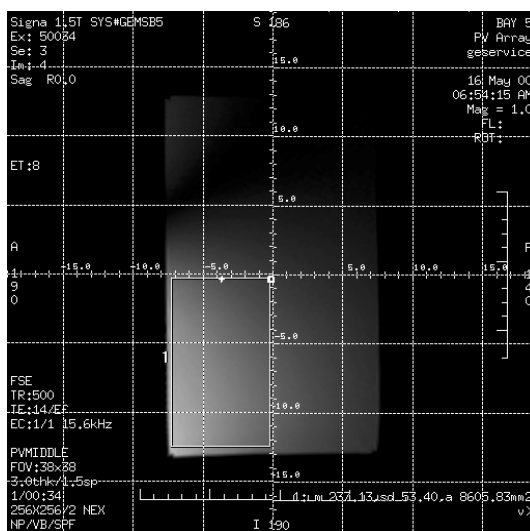


Figure 15: PV Upper/Middle/Lower Receiver 4.
ROI $\approx 8600 \text{ mm}^2$.
7.5 cm A-P, 12.5 cm S-I.

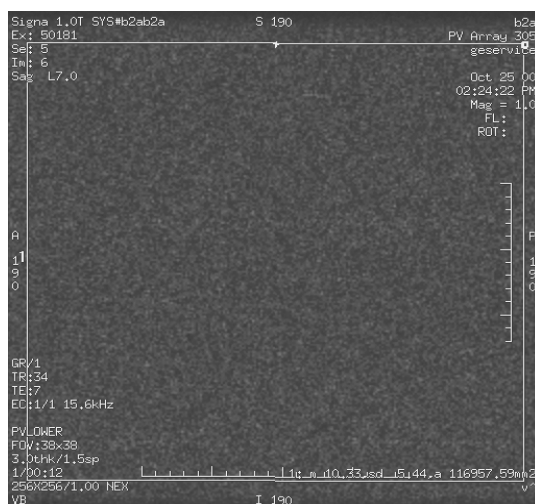


Figure 16: PV Upper/Middle/Lower noise scan.
ROI ≈ 118000 .

Individual Element Performance

Figures 17 through 21 below show typical receiver images for PV Upper, Middle, or Lower configurations.

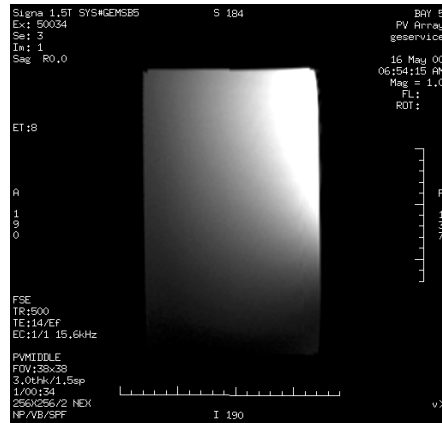


Figure 17: PV Upper/Middle/Lower receiver image #1.

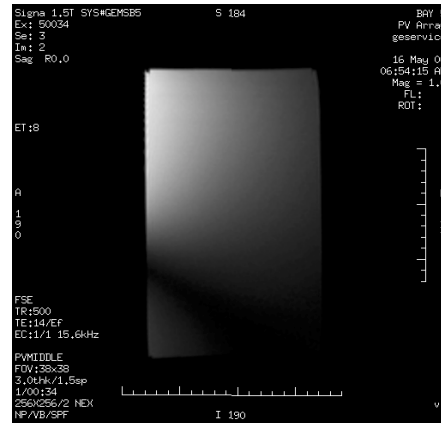


Figure 18: PV Upper/Middle/Lower receiver image #2.

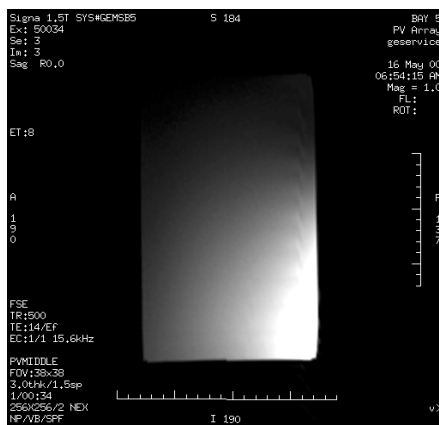


Figure 19: PV Upper/Middle/Lower receiver image #3.

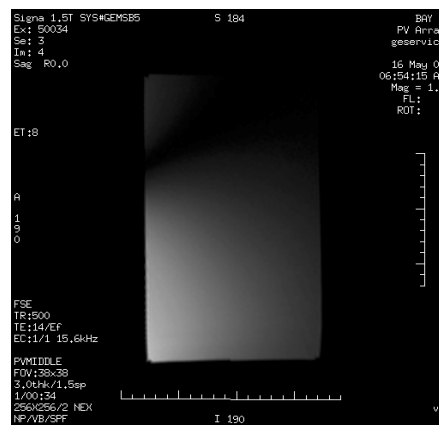


Figure 20: PV Upper/Middle/Lower receiver image #4.

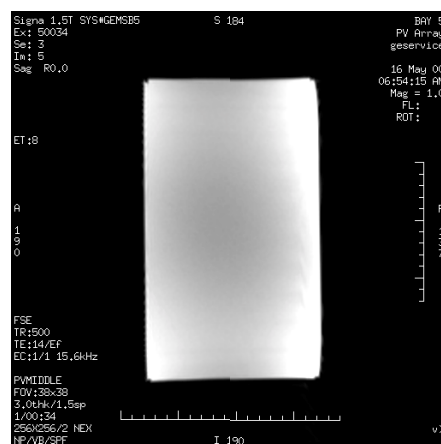


Figure 21: PV Upper/Middle/Lower composite image.

SNR Specification

The SNR measurements must be greater than or equal to the following specifications:

SNR SPECIFICATIONS – TABLE 3-2-5

	CH0	CH1	CH2	CH3
PVUPPER	60.2	58.3	77.3	63.3
PVMIDDLE	97.8	82.0	110.8	65.4
PVLOWER	126.8	82.2	152.2	76.9

3-3 External Cable Check

Check for continuity in the output cable.

Step 1 – Using a Digital Volt Meter (DVM), separate the output cable from the anterior coil by pushing the buttons located on the sides of the cable housing and gently pulling the connector out of the socket.

Step 2 – Follow the same procedure for the posterior coil. Make sure there is continuity between the appropriate pin at the 30-pin connector that plugs into the system and the connectors that plug into the anterior and posterior coils

Step 3 – Refer to *Figure 22* and Center Pin and Ground Connections – Table 3-3 for the correct pins.

CENTER PIN AND GROUND CONNECTIONS – TABLE 3-3

Connector to Coil Pins	30-Pin Connector to Cradle
Anterior, Center Pin 1	Pin B11
Anterior, Ground 1	Pin A11
Anterior, Center Pin 2	Pin B15
Anterior, Ground 2	Pin A15
Anterior, Center Pin 3	Pin B3
Anterior, Ground 3	Pin A3
Anterior, Center Pin 4	Pin B7
Posterior, Center Pin 1	Pin A7
Posterior, Ground 1	Pin B5
Posterior, Center Pin 2	Pin A5
Posterior, Ground 2	Pin B1
Posterior, Center Pin 3	Pin B13
Posterior, Ground 3	Pin A13
Posterior, Center Pin 4	Pin B9
Posterior, Ground 4	Pin A9

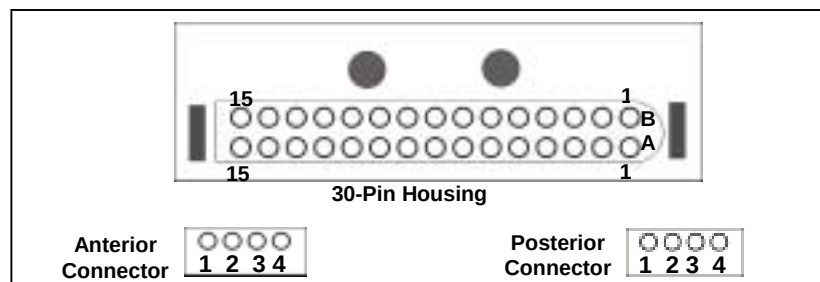


Figure 22: Pin layouts of 30-pin housing and anterior and posterior connectors.

3-4 PIN Diodes Check

Step 1 – Remove the coil covers.

Step 2 – Gently pry up the four white screw covers located down the middle of the anterior cover and the three white screw covers located down the middle of the posterior cover.

Step 3 – Remove all the screws found under the covers.

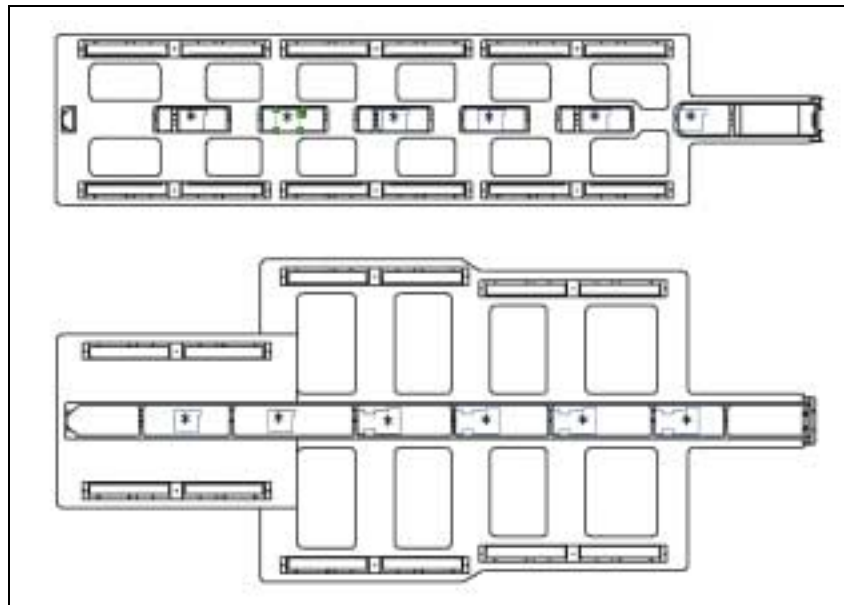
Step 4 – Remove the center covers from both the anterior and posterior coils.

Step 5 – Check the pin diodes located on the feed boards of the individual coil elements with a DVM. Every coil element has one diode (see *Figure 23* for the location of the diodes).

Step 6 – If a diode is found to be either open or shorted:

- Take note of the diode's cathode-anode orientation.
- Carefully de-solder the surface mounted diodes from the feed board.
- Cut the leads of a new diode to 1/8".
- Place the new diode into the hole on the PCB. Align the leads of the diode with the surface mount pads.

The cathode pad is designated with a dot near the pad (see *Figure 24* for diode mounting).



*Figure 23: Pin diode placement for both the anterior and posterior coils
(* represents pin diode location).*

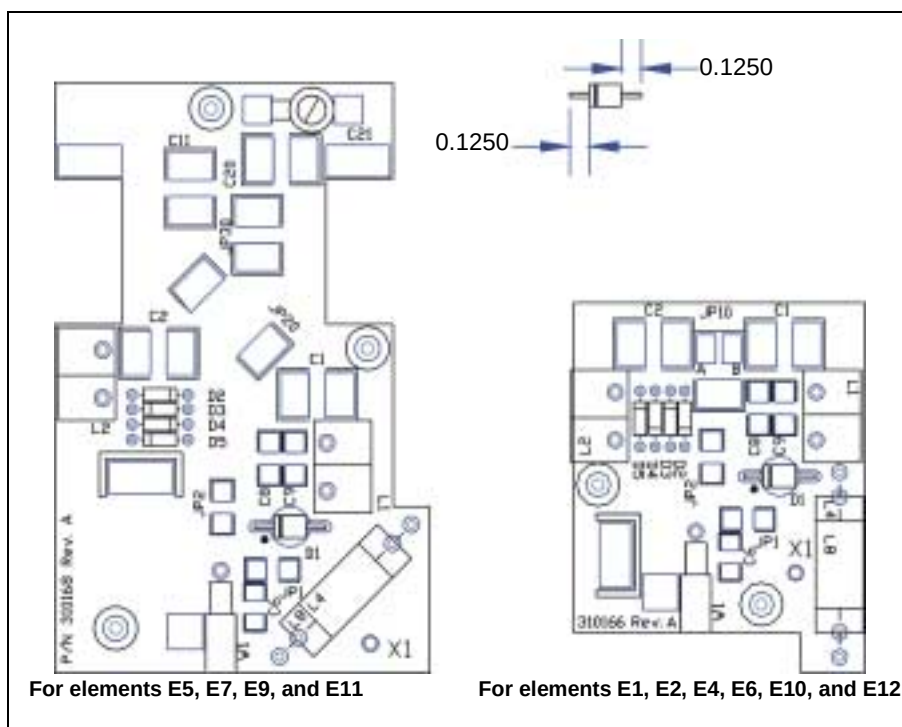


Figure 24: Diode mounting.

3-5 Mechanical Hardware Check

None.

3-6 Troubleshooting Tips

When troubleshooting the coil, refer to the receiver images for PV Upper, Middle, or Lower configurations in *Figures 17 through 21* in Section 3-2-5 SNR Image Analysis and the element-to-receiver mapping diagrams in *Figures 6, 7 and 8*.

Symptom #1: The system reports a coil fault during prescan or does not recognize the coil connection to the system when selected in the software.

Probable Cause	Suggested Actions	Resolution
The coil connector has become disconnected from the system interface.	Check to make sure the coil connector is fully engaged.	Engage connector and try the scan again. Make sure the indicator light on the patient table turns green with the proper coil selected on the console.
One or more PIN diodes have failed.	From inside the coil, check the PIN diodes using a DVM. Use diode mode for forward-bias measurements and resistance mode for reverse-bias measurements. Note that there should be no reverse-bias leakage. Bad PIN diodes are usually shorted or exhibit reverse-bias leakage current.	Replace the coil.
There is a DC short or open somewhere within the DC path inside the coil.	Check the coil and associated DC paths with a DVM, in both diode check and resistance modes.	If a short or open is observed, check the output cable by itself (see below). If the cable and PIN diodes are okay, then replace the coil.
The output cable has a short or an open.	Disconnect the cable at the coil and check each coaxial line with a DVM (resistance). Try moving and twisting the cable as you watch the meter to find any intermittent connections. Inspect the center pins of the SMB and Bendix connections for wear and proper engagement into the mating connection.	Replace the coil.
There is a problem with the DC bias from the system interface or the coil ID circuitry.	Check to ensure the correct DC bias is being provided to the coil during transmit and receive per GEMS System Service Manual. Try a similar 2-channel coil to see if the same problem occurs	Correct MRI system problem.

Symptom #2: The coil does not pass SNR tests or exhibits poor image quality on patient scans.

Probable Cause	Suggested Actions	Resolution
One or more of the coil channels has a high noise level.	Look at the uncombined images to determine which channel has the problem. Compare the noise standard deviation measurements between channels and system performance logs; they should be approximately the same. Try swapping preamplifiers between channels to see if the problem follows the preamp. If it does, then the preamplifier is bad.	Replace the coil.
One or more of the coil channels has low signal.	Look at the uncombined images to determine which channel has the problem. Compare signal mean measurements between channels and system performance logs.	Replace the coil.
Excessive ghosting is causing the noise standard deviation measurements to be artificially high.	Window and level the images down to look at the background for signs of ghosting. Try padding the phantom in the holder. Try running the scan without the phantom holder to see if the ghosting diminishes. Try turning off the cold heads to see if the ghosting is vibration-related.	If padding or changing the phantom position can minimize ghosting, then the problem is caused by excessive vibration elsewhere in the system. If the ghosting is not symptomatic of phantom positioning or padding, then replace the coil.
There is a problem with the DC bias from the system interface or the coil ID circuitry.	Check to ensure the correct DC bias is being provided to the coils during transmit and receive per GEMS System Service Manual.	Correct MRI system DC bias problem.

SECTION 4 – MAINTENANCE

4-1 Coil Care



Detach the coil connector from the scanner before attempting to clean the coil. Do not touch the connectors with bare fingers. Never press a sharp object against the surface of the connector. Do not reattach the connector after cleaning the coil until the coil has dried completely. Electric shock may result if the coil is attached to the system during cleaning or when it is wet.



Do not spray or pour the cleaning solution directly on the coil. Do not submerge the coil in the solution. The coil contains sensitive electronics components that could be damaged by the solution. The coil cannot be sterilized and should be cleaned only according to the procedure outlined in this section.

The following solutions are recommended for the coil and pad surfaces: (1) a ten percent bleach solution (some discoloration may occur), (2) one ounce commercial dishwashing liquid mixed with one gallon of water or (3) warm water. Apply cleaning solution to a soft cotton cloth and proceed to clean. The user place a cotton sheet over the coil before positioning the patient to prevent soiling of the coil. If the coil is soiled, clean the coil as described above.

4-2 Special Care Requirements

Prior to returning the coil for service, use a ten percent bleach solution (as described above) to eliminate risk of exposure to potentially infectious materials.

SECTION 5 – REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

5-1 External Cable Replacement

The cable is a separate part of the coil. If the cable needs to be replaced, detach it from the coil and replace it with a new cable assembly.

5-2 Mechanical Hardware Replacement

None.

SECTION 6 – RENEWAL PARTS

6-1 Field Replaceable Units

FIELD REPLACEABLE UNITS – TABLE 6-1

Part Name	GE Part #	USAI Part #
Coil	2285178-2	100098
Cable Assembly	2285178-3	110036
SNR Phantom	2192406-6	150034
Phantom Strap	2285178-4	150171
Loading Phantom	2192406-7	150035
Phantom Positioner Set	2192406-9	150036

6-2 Other Replaceable Accessories

OTHER REPLACEABLE ACCESSORIES LIST – TABLE 6-2

Part Name	GE Part #	USAI Part #
Ramp Pad	E8800ZT	150040
Patient Comfort Pad	E8800ZW	150075
Foam Spacer Block	E8800ZY	150041
Velcro Strap Set	E8800ZZ	150042
Coil Shipping Container Set		150277
Phantom Shipping Container Set		150278

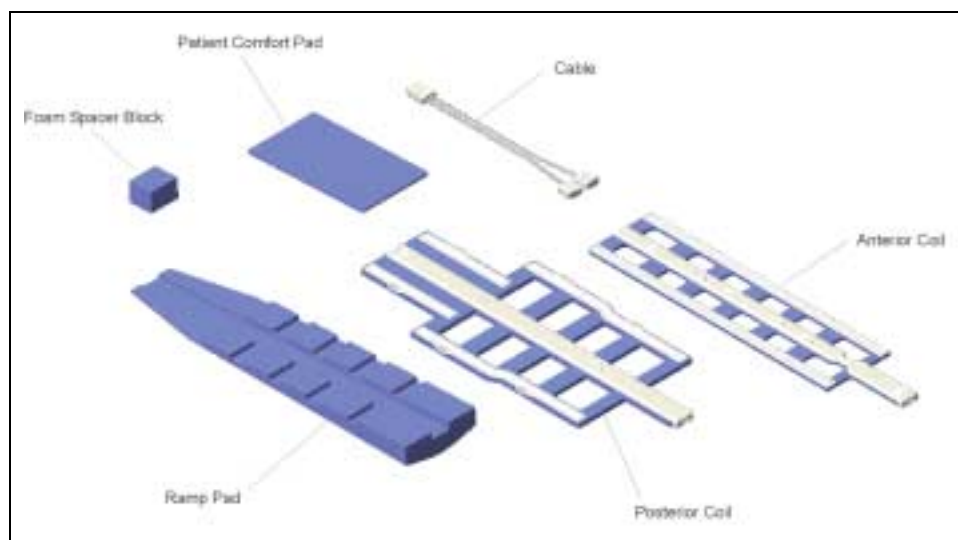


Figure 25: Parts of the coil.

SECTION 7 – APPENDIX

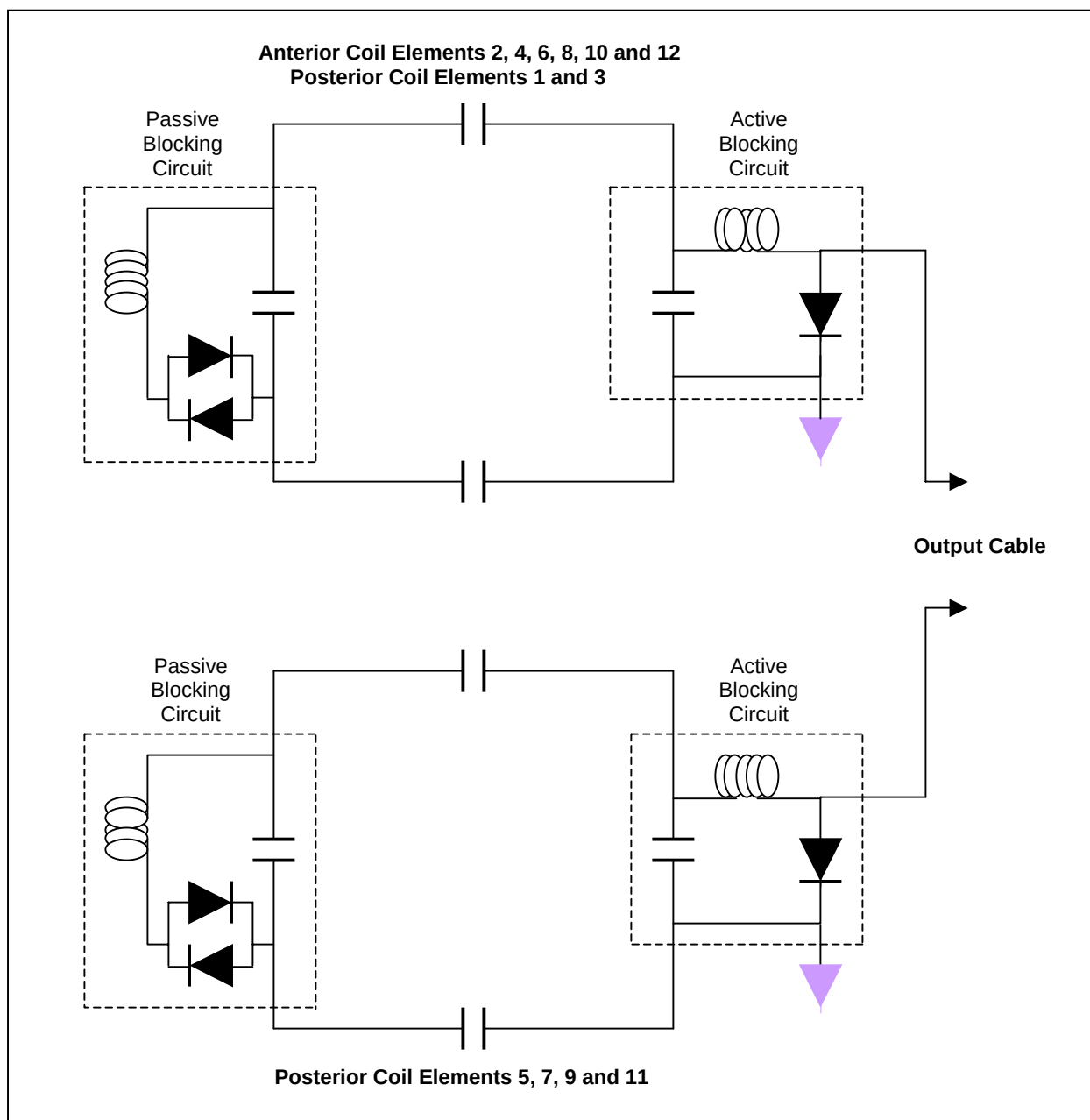
7-1 SNR Data Sheet

Use the table provided below to record the calculated signal to noise ratio (SNR) data obtained from the Functional Checks section.

Date	Comments							
Mode	R1	R2	TG	Element	Signal Mean	Noise Std Dev	SNR	Spec Limit
PVUPPER				0				60.2
				1				58.3
				2				77.3
				3				63.3
PVMIDDLE				0				97.8
				1				60.2
				2				58.3
				3				77.3
PVLOWER				0				126.8
				1				82.2
				2				152.2
				3				76.9

Date	Comments							
Mode	R1	R2	TG	Element	Signal Mean	Noise Std Dev	SNR	Spec Limit
PVUPPER				0				60.2
				1				58.3
				2				77.3
				3				63.3
PVMIDDLE				0				97.8
				1				60.2
				2				58.3
				3				77.3
PVLOWER				0				126.8
				1				82.2
				2				152.2
				3				76.9

7-2 Schematic



7-3 Coil Configuration

Parameter	Modes		
Coil Name	PVUPPER	PVMIDDLE	PVLOWER
Coil Type	3	3	3
Extremity Coil	no	no	no
Cable Loss	1.24	1.24	1.24
Coil Loss	0.549	0.549	0.549
Recon Scale Factor	1	1	1
Linear vs. Quadrature	1	1	1
Multiple Receiver Coil?	yes	yes	yes
Number of Receivers	4	4	4
Starting Receiver ID	0	0	0
Ending Receiver ID	3	3	3
Multi-Coil Port Enable	3	12	9
Multi-Coil Port Error Enable	3	12	9
Additional Transmit Attenuation	0	0	0
Number of Fast Receivers	0	0	0
Starting Fast Receiver ID	4	4	4
Ending Fast Receiver ID	4	4	4
Start TA Value	90	90	90
Start RG Value	12	12	12
Multi-Coil Recon Enable	0	0	0
Head Default Freq Direction	0	0	0
SCIC Axial	1.5 0.35 0.2 5.0 128 0.2 1.05	0	0
SCIC Sagittal	1.5 0.35 0.2 5.0 128 0.2 1.05	0	0
SCIC Coronal	1.5 0.35 0.2 5.0 128 0.2 1.05	0	0

REVISION HISTORY

Rev	Date	Author	Primary Reason for Change
A	06/11/01	Reva Zaretsky	First Issue
B	06/21/01	Reva Zaretsky	Revised Tools Required Table